

CO5-AT2-Q3:

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Course Name: DAA for Complexity Analysis

Set Cover Approximation with Maximum Uncovered Gain Problem Understanding

Set Cover problem selects subsets from a collection to cover all elements of a universe. The greedy approximation chooses the subset that covers maximum uncovered elements.

Algorithm Used

Greedy Approximation Algorithm

Constraints

- All elements must be covered.
- Selected subsets should minimize number of choices.
- Maximum uncovered gain should be selected.

Solution Approach

Steps:

Initialize uncovered elements.

Calculate gain for each subset.

Select subset with maximum gain.

Update covered elements.

Repeat until all elements are covered.

Correctness

The algorithm ensures every element is covered by continuously selecting subsets with highest contribution.

Complexity Analysis

Time Complexity:

$O(m \times n)$

Space Complexity:

$O(n)$

Conclusion

Greedy Set Cover provides an efficient approximation solution for large optimization problems where finding the exact solution is computationally expensive.